Kolsky

January 26, 1955

Memorandam for file:

Dr. Harwood Kolsky of the Los Alamos Scientific Laboratory discussed with me on this date some of the future requirements for large scale scientific problems such as are contemplated at Los Alamos.

At present, the 701 is being used on 12 x 12 point networks. He forsees 25 x 25 in the near future and thinks that 150 x 150 would be a future goal. = 250,000 words

On the 701, about 600 order executions are required per mesh point, taking about 75 milliseconds. Approximately 25% of these orders are for address modification and 10 - 15% are multiply.

Storage requirements are now 7 words per mesh point. He would like to increase this to 8 or 9 words.

For future requirements, an n x n system will require then, approximately 10 n^2 words of storage and k n^3 units of time for the problem, where k is the amount of time taken for a single point.

However, if another space dimension were added, the memory requirements will be related to n^3 and the calculation time to n^4.

Dr. Kolsky believes that this type problem may be solved by dealing with data in blocks rather than on a completely random basis. In this case, if fast tapes were available to fill memory, the requirement would be that the block being operated upon plus the immediately adjacent blocks be in memory at one time in order that diffusion between blocks could be handled.

Dr. Kolsky expressed an interest in some device similar to the wire capsules which were used with the SEAC. These would be useful in a library of partialy solved problems which will, at some future time, be continued. He finds binary cards useful for this purpose at present, and feels that large reels of magnetic tape would not be useful since he would like to store only one problem per storage package and the storage requirements are now only 90 binary cards.

Dr. Kolsky believes that future machines should have

1 Decimal number system
2 Simultaneous read-write-compute
3 Self checking.

L. C. Hubbard

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